Digital Circuits Lab Report

LAB 1: CMOS Logic Circuit

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1) Aim

To design a CMOS logic circuit that generates a warning signal based on temperature and humidity sensor outputs, and to obtain both active-low and active-high warning signals.

2) Summary of the Experiment

In this experiment, two sensors—temperature and humidity—are considered. Each sensor produces a HIGH output when its value exceeds the acceptable limit.
A CMOS logic circuit is designed such that a warning is generated only when both sensors exceed their limits. First, an active-low warning signal is obtained, and then it is converted into an active-high warning signal using an inverter.

3) Components Used

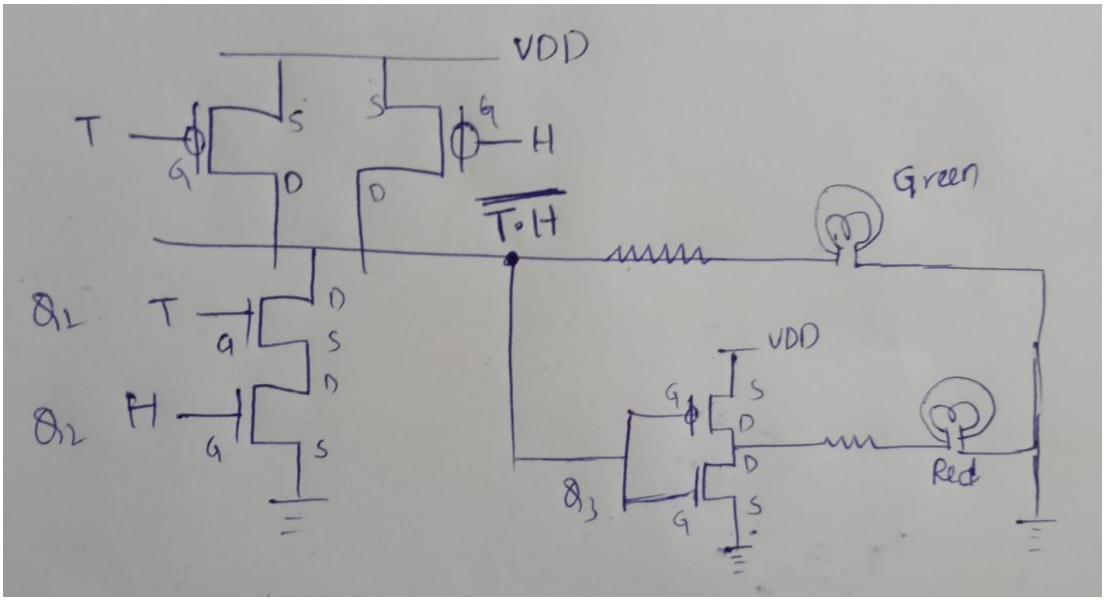
1. HCF4007UBE IC
2. LEDs (RED and GREEN)
3. Power supply and connecting wires
4. Bread board
5. Resistors

5) Truth Table

T	H	
1	1	red
1	0	green
0	1	green
0	0	green

6) Circuit Description

- The CMOS NAND gate is implemented using PMOS transistors in parallel and NMOS transistors in series.
- The inverter is implemented using one PMOS and one NMOS transistor.
- Green LED indicates safe condition (active-low).
- Red LED indicates warning condition (active-high).



8) Conclusion

The required CMOS logic circuit for a smart home warning system was designed. Both active-low and active-high warning signals were observed.